

CRYSTALLINE FORMS OF DANUGLIPRON

CROSS REFERENCE

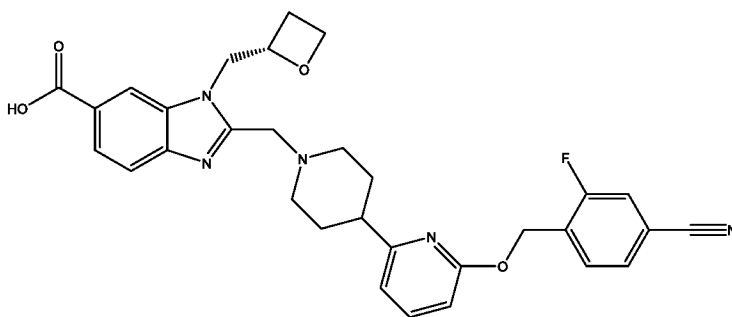
5 This application claims the priority of our Indian provisional applications 202341031510 filed on 03rd May 2023 and 202341063459 filed on 21st September 2023.

FIELD OF THE INVENTION

The present application relates to crystalline forms of Danuglipron and
10 processes for preparation thereof. Specifically, the present application relates to crystalline forms of DN1, DN2, DN3 of Danuglipron and processes for preparation thereof. The present application also relates to use of crystalline forms of Danuglipron for the preparation of a medicament for treatment and/or prevention of Type 2 Diabetes Mellitus, pre-diabetes, obesity, NASH, and
15 cardiovascular diseases.

BACKGROUND OF THE INVENTION

Danuglipron is the adopted name for a drug chemically described as 2-[(4-{6-[(4-cyano-2-fluorophenyl)methoxy]pyridin-2-yl}piperidin-1-yl)methyl]-1-[(2
20 S)-oxetan-2-yl)methyl]-1H-benzimidazole-6-carboxylic acid and is represented by structural Formula I.



Formula I

Danuglipron is currently under clinical investigation for the treatment of
25 type 2 diabetes mellitus and obesity.